

Predicting Song Popularity Using Supervised Machine Learning: An Analysis of the Almost Million Songs Dataset 2025

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Abstract

The vast amounts of data that are being generated by music streaming services like Spotify in recent years have been the subject of growing interest. Many of these vast archives of audio data not only represent a treasure trove for music lovers but can also be analyzed using complex statistics to help understand and predict factors that influence a song's popularity. Numerical audio features of music streaming services such as Spotify for the assessment of factors influencing song popularity. This work investigates the ability of numerical audio features to predict the popularity of songs by using supervised machine learning for regression. The data set used in this work is the Almost Million Songs Dataset 2025, that comprises 899,702 tracks from the Spotify database, each song has 16 attributes. 11 of these attributes are continuous and were retrieved from the Spotify API. 899,702 tracks and 16 attributes per song, 11 of them being continuous audio features, which were retrieved from the Spotify API.

The following regression models have been evaluated for the task of predicting song popularity: Linear Regression, Decision Tree regression, Random Forest regression, K-Nearest Neighbors (KNN) regression and Multi-Layer Perceptron (MLP) regression. Multi-Layer Perceptron (MLP). Among all the models we have tested, the Random Forest model has achieved the best results in terms of the previous metrics with an R^2 score of 0.141, an RMSE of 16.53, and an MAE of 13.61. 0.141, RMSE of 16.53, and MAE of 13.61.

The results demonstrate that audio features contain statistically significant information related to song popularity; however, audio features alone are not enough to explain the popularity of songs. Other relevant signals, for example lyrics, artist metadata or even metadata and social interaction data of users, should be integrated as well in order to achieve better results. social interaction data may improve predictive performance.

Index Terms

Keywords: music popularity prediction, supervised learning, random forest, audio features, regression, machine learning

I. INTRODUCTION

The global music streaming industry has grown rapidly over the last few years and today reaches hundreds of millions of subscribers. On top of local storage of songs on users' devices, most music streaming services rely on online infrastructure to deliver music to their users. Large amounts of data are generated along the way, both on users' behavior and on the audio being played. the global music streaming industry, with hundreds of millions of subscribers accessing music and millions of tracks to listen to on services like Spotify, Apple Music, and YouTube Music. large amounts of audio-related data.

Audio features extracted from songs, such as tempo, energy, danceability, acousticness, and valence, can provide objective descriptions of musical characteristics. These features are useful for improving recommendation systems, playlist generation, music marketing, and early identification of successful songs. Traditionally, predicting hit songs relied heavily on expert knowledge, radio airplay statistics, and chart history. However, the availability of large-scale music APIs to analyze songs in detail to try to predict their popularity before they even released to the public. audio features. Can numerical audio features be used to predict song popularity using supervised regression? This paper investigates this question by evaluating five regression models (Linear Regression, Decision Tree, Random Forest, K-Nearest Neighbors and Multi-Layer Perceptron) on a set of audio features that have been used before in the field of music recommendation. In this work, five supervised regression methods for hit song prediction were tested. The following methods were compared: Linear Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN) and Multi-Layer Perceptron (MLP). Forest, K-Nearest Neighbours (KNN), and Multi-Layer Perceptron (MLP).

A. Related Work

Pachet and Roy (2008) demonstrated that musical features alone have limited but statistically significant predictive power for measuring a song's potential to become a hit. for chart success. Ni et al. (2011) explored hit song classification using audio, lyrical, and social features. Dhanaraj and Logan (2005) used Support Vector Machines (SVMs) with acoustic features to predict hit songs. (2005) used Support Vector Machines (SVMs) with acoustic features to predict hit songs. Mart´ın-Guti´errez, Guti´errez-Arriola, Guti´errez-Mart´ın (2020) (in Spanish) used gradient boosting and neural networks to predict Spotify track features, again finding that ensemble models performed much better than linear baselines. features and found that ensemble models consistently outperformed linear baselines. As was already mentioned in the related work section, audio features explain only a small part of the popularity of songs. Other external factors, like the reputation of the artist, the time of release, social media exposure and other marketing measures play a much bigger role for the popularity of songs. external factors such as artist reputation, release time, social media exposure, marketing etc. are also playing a role in a song's popularity.

II. DATASET DESCRIPTION

A. Overview

The Almost Million Songs Dataset 2025 was used. The public dataset provided on Kaggle includes 899,702 tracks, including 16 attributes and 4 different categories that relate to the tracks (e.g. track information, popularity, genres). 11 of the 16 categories are used as continuous audio features that were continuously provided by the Spotify API for the songs' tracks. This data set includes 899,702 tracks in total and contains 16 attributes for each track. These attributes include track information, its popularity as well as the genres to which the track belongs. The audio features of the tracks are based on 11 continuous features that have been extracted from the Spotify API.

B. Features

Table I summarises the audio features used as predictors.

TABLE I
AUDIO FEATURES USED AS PREDICTORS

Feature	Description
Tempo	Estimated beats per minute (BPM)
Energy	Perceptual measure of intensity (0–1)
Key	Musical key (0 = C to 11 = B)
Mode	Major (1) or minor (0) modality
Time Signature	Estimated beats per bar
Speechiness	Presence of spoken words
Danceability	Suitability for dancing
Valence	Musical positivity
Acousticness	Confidence of acoustic track
Liveness	Presence of live audience
Instrumentalness	Likelihood of no vocals

C. Preprocessing

Rows containing missing values were removed from the data set. Songs with a score for popularity of zero were removed as well. The resulting data set consisted of 670462 songs. were excluded. After preprocessing, the working dataset contained 670,462 tracks. The popularity target variable had a mean of 29.3 and a standard deviation of 17.8, so there is a wide range of values in this variable. popularity scores.

III. PROBLEM STATEMENT AND RESEARCH QUESTIONS

A. Problem Statement

Given a set of numerical audio features for a song, how to predict the song's popularity score on a continuous scale from 1 to 100. Fig 2: Continuous popularity score between 1 and 100. We're looking for a supervised regression solution. Let $x \in \mathbb{R}^{11}$ be the feature vector of a song and $y \in \mathbb{R}$ its corresponding popularity score. We are searching for a function f that, given the feature vector x of a song, returns an approximation $f(x)$ of its corresponding popularity score y is produced. We are searching for a function f that minimises the expected squared error: $E[(y - f(x))^2]^2$.

$f : \mathbb{R}^{11} \rightarrow \mathbb{R}$

that minimises the expected squared error: $E[(y - f(x))^2]$

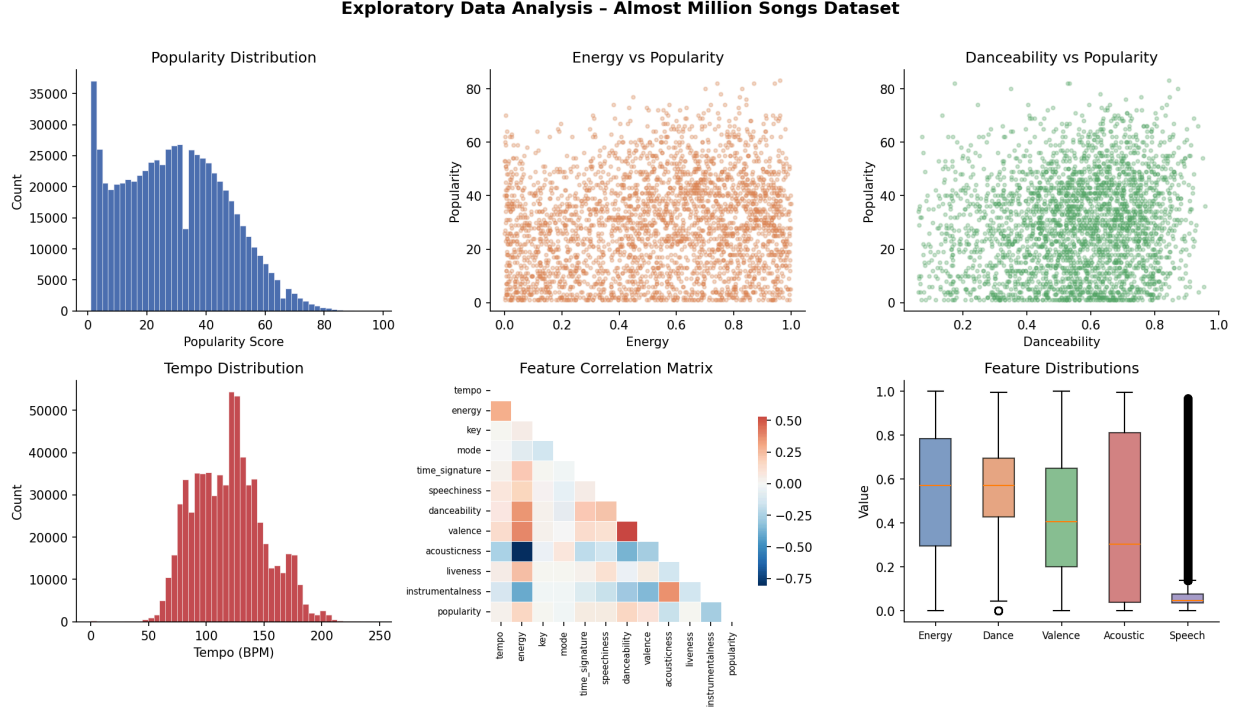


Fig. 1. Exploratory data analysis including popularity distribution, feature distributions, and feature correlations.

B. Research Questions

- RQ1: Which audio features correlate most strongly with song popularity?
- RQ2: Which supervised regression model best predicts popularity?
- RQ3: How does model complexity affect predictive performance?
- RQ4: What are the limitations of audio-feature-based popularity prediction?

IV. METHODOLOGY

A. Experimental Setup

Due to computational limitations, a random sample of 100,000 songs was selected from the preprocessed dataset. The data were split into training (80%) and testing (20%) sets. We standardize all features of the songs in the data set using a StandardScaler in order to train the models on the data set afterwards.

B. Models

LR - Linear Regression: Parametric model that predicts the popularity using a weighted sum of the song's audio features.

Decision Tree (DT): a tree-based non-linear model with maximum depth of 8 in order to prevent overfitting.

Random Forest (RF): This is a combination of 100 decision trees with $max_depth = 10$ (i.e. max number of leaf nodes), where trees are

K-Nearest Neighbours (KNN): An instance-based model that for every song in the test set calculates the mean popularity of the 10 closest songs in the training set.

Multi-Layer Perceptron (MLP): a feedforward neural network consisting of one hidden layer of 128 ReLU units followed by another hidden layer of 64 ReLU units. We used Adam for optimization.

C. Evaluation Metrics

The following regression metrics were used:

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 (Coefficient of Determination)

V. RESULTS AND DISCUSSION

A. Feature Correlation Analysis (RQ1)

Instrumentalness and acousticness are the two audio features with the strongest negative correlation with popularity. The strongest positive correlation with popularity is found for danceability. The strongest positive correlation was found for danceability, whilst instrumentalness and acousticness showed the strongest negative correlations with popularity. The absolute correlation values, however, were low, supporting the use of non-linear models.

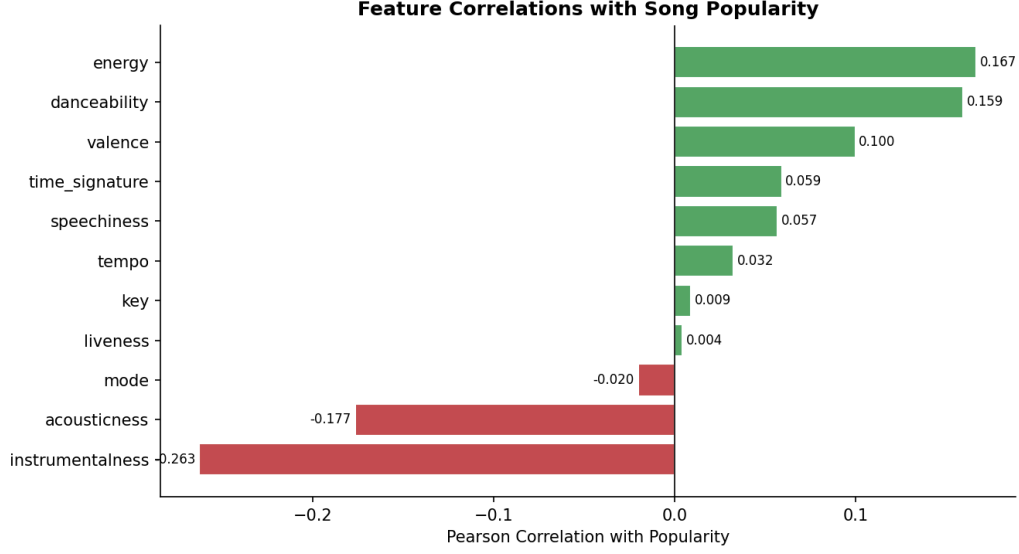


Fig. 2. Pearson correlations between audio features and song popularity.

B. Model Performance Comparison (RQ2, RQ3)

TABLE II
MODEL PERFORMANCE ON THE TEST SET

Model	RMSE	MAE	R^2
Random Forest	16.53	13.61	0.141
MLP	16.60	13.68	0.133
Decision Tree	16.75	13.76	0.118
Linear Regression	17.07	14.12	0.083
KNN	17.43	14.23	0.044

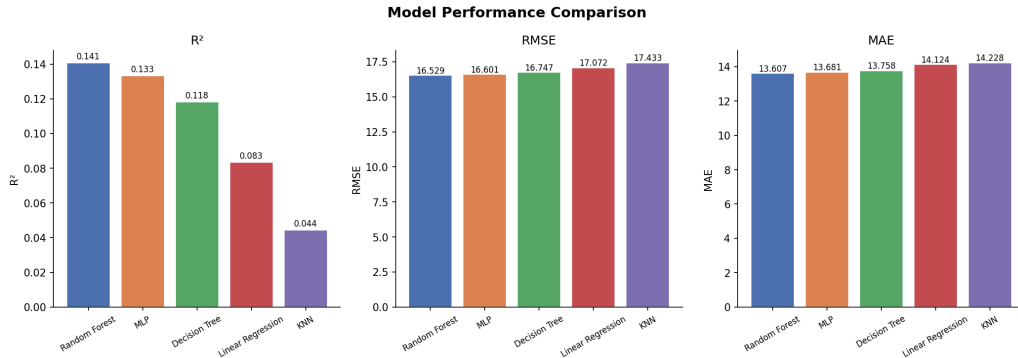


Fig. 3. Comparison of all models across RMSE, MAE, and R^2 .

C. Performance Analysis

Random Forest achieved the highest score for the quality of predictions with the result of $R^2 = 0.141$. This model is able to efficiently forecast interactions of great number of features, non-linear relationships while reducing variance. MLP achieved the second-best performance, while KNN performed worst due to the curse of dimensionality affecting distance-based learning methods.

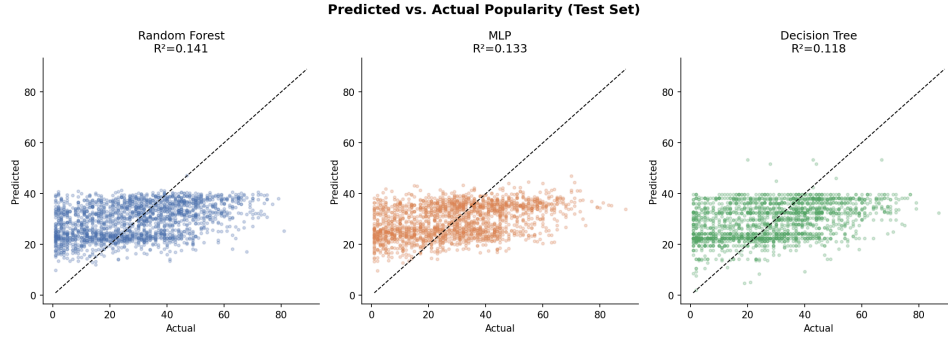


Fig. 4. Predicted vs. actual popularity for the top-performing models.

D. Feature Importance

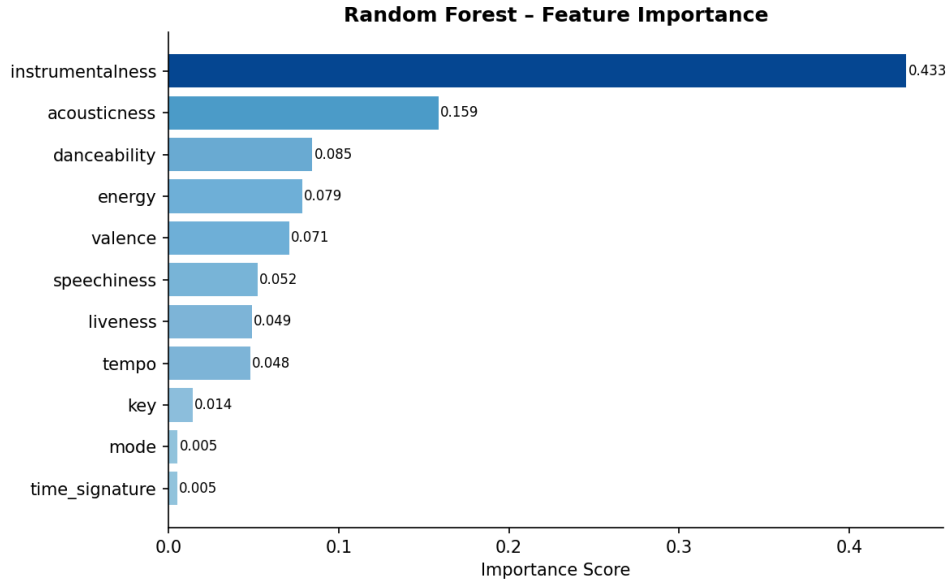


Fig. 5. Random Forest feature importance scores.

Feature importance of the Random Forest model showed that the model used mainly instrumentalness, acousticness, danceability and energy in its model. informative features.

E. Residual Analysis

The residual distributions are centred approximately around zero which means that the models are making largely unbiased predictions, even though a large part of the variance in the data remains unexplained by the models. stantial unexplained variance remains.

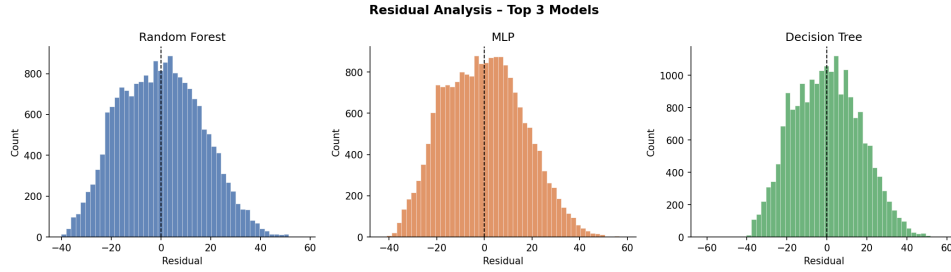


Fig. 6. Residual distributions for the top-performing models.

F. Limitations (RQ4)

While the audio features of songs contain predictive information for popularity, there is not enough information to accurately predict the popularity of songs. Other factors such as the artist, marketing, time of release, social media, and even the lyrics of songs strongly influence the popularity of songs. There are other factors, such as the artist's reputation, the song's marketing, the song's release time, the artist's social media presence, etc., that affect a song's popularity significantly more than the audio features. Future work could use additional multi-modal data such as song lyrics, artist data, social interaction and release in time with respective temporal release context. release-context information.

VI. CONCLUSION

Supervised machine learning for song popularity prediction using audio features from the Almost Million Songs Dataset 2025.

The best performing model in this study is the Random Forest model, followed by Support Vector Regressor, Decision Trees, Support Vector Machine, Neural Network and Linear Regression models.

features were instrumentality, acousticness, danceability, and energy.

The results obtained in this work show that the audio features used in this work contain statistically significant information regarding popularity; however, they only explain a small portion of the total popularity variance.

explain only a small proportion of total popularity variance. Future work will be done to integrate more multi-modal data sources and implement more advanced deep learning architectures in order to improve the model's performance. predictive performance.

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